Lets revise some formulae:

$a^2 + b^2 =$
$a^3 + b^3 =$
$a^4 + b^4 =$
$a^5 + b^5 =$
$a^6 + b^6 =$
$a^7 + b^7 =$
$a^8 + b^8 =$

If ‘a’ and ‘ b’ are the roots of the equation $x^2 - 14x + 11 = 0$, find $a^2 + b^2$?

a) 174 b) 196 c) 218 d) None

If α and β are the roots of the equation $x^2 - 6x + 6 = 0$, then what is $\alpha^3 + \beta^3 + \alpha^2 + \beta^2 + \alpha + \beta$ equal to ?

a) 150 b) 138 c) 128 d) 228

a and b are the roots of the quadratic equation $x^2 - x - 1 = 0$ what is the value of $a^8 + b^8$?

a) 47 b) 54 c) 59 d) 68

If α and β are the roots of the quadratic equation $ax^2 + bx + c = 0$, then the value of $\frac{\alpha^2}{\beta} + \frac{\beta^2}{\alpha}$ is:

a) $\frac{3bc-a}{b^2c}$ b) $\frac{3abc-b}{a^2c}$ c) $\frac{3abc+b}{a^2c}$ d) $\frac{ab-b\ c}{2b^2c}$

α β



If a and b are the roots of equation $x^2 - x + 1 = 0$, then which equation will have roots $a^3 \, and \, b^3$?

a) $x^2 + 2x + 1 = 0$

b) $x^2 - 2x - 1 = 0$

c) $x^2 + 3x - 1 = 0$

d) $x^2 - 3x + 1 = 0$

In solving a problem, one student makes a mistake in the coefficient of the first degree term and obtains -9 and -1 for the roots. Another student makes a mistake in the constant term of the equation and obtains 8 and 2 for the roots. The correct equation was

a) $x + 10x + 9 = 0$

b) $x^2 + 10x - 9 = 0$

c) $x^2 - 10x + 9 = 0$

d) None of these

If the equations $x^2 + 5x + 6 = 0$ and $x^2 + kx + 1 = 0$ have a common root, then what is the value of k ?

a) $-\frac{5}{2}$ or $-\frac{10}{3}$ b) $\frac{5}{2}$ or $\frac{10}{3}$ c) $\frac{5}{2}$ or $-\frac{10}{3}$ d) $-\frac{5}{2}$ or $\frac{10}{3}$

If $sin\theta$ and $cos\theta$ are the roots of the equation $ax^2 - bx + c = 0$, then which one of the following is correct ?

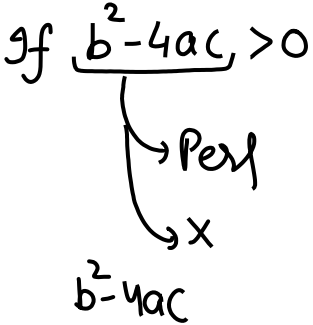
a) $a + b + 2ac = 0$

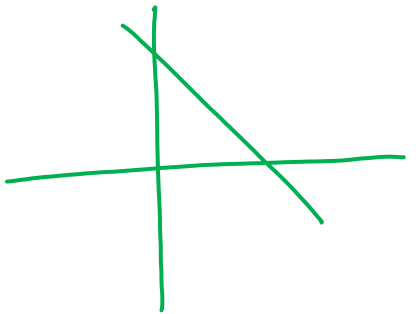
b) $a - b + 2ac = 0$

c) $a^2 + c^2 + 2ab = 0$

d) $a^2 - b^2 - 2ac = 0$

Nature of roots:



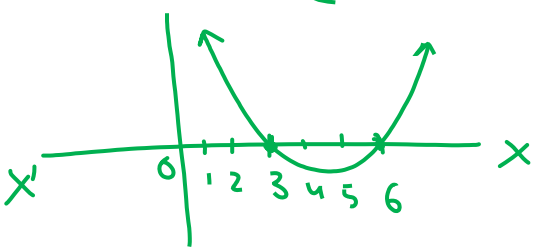


Conjugate Pair

$ax^2+bx+c=0$

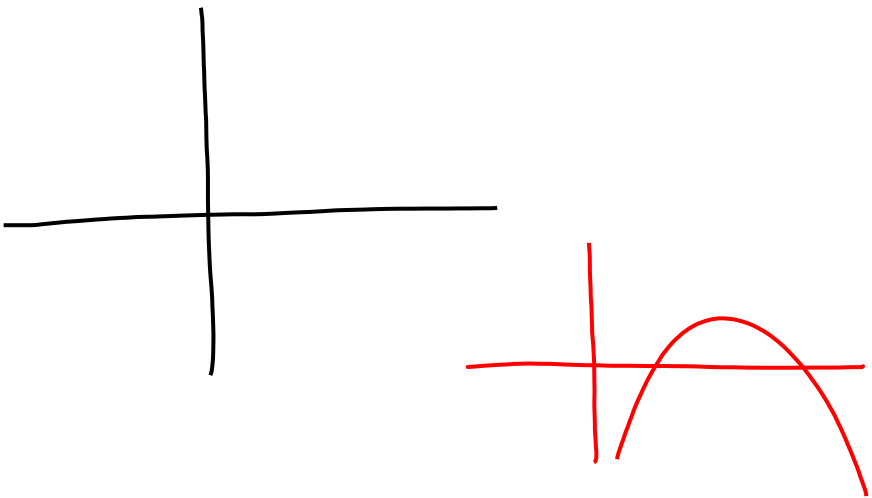
$3-\sqrt{5}$, $3+\sqrt{5}$ → if $a,b,c \in$ Rational

$2+3i$, $2-3i$ → if $a,b,c \in$ Real.

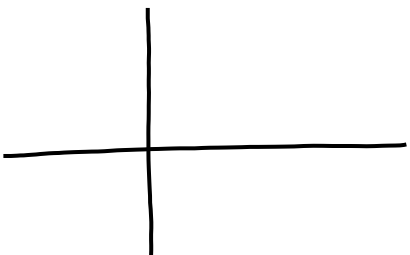


$x^2-9x+18=0$

Conjugate pairs ( ):



Maximum / minimum value:



At what value of p, the expression $p^2 - 4p + 5$ attains its minimum?

a) 2 **b) 3** **c) 4** **d) – 2**

At what value of x, does the expression $2x^2 - 48x + 3$ attains it minimum?

a) 24 **b) 12** **c) 6** **d) – 12**

1 The expression $(x-3)(x-7)$ attains its minimum value when $x =$

a) 5 b) -5 c) 10 d) -15

② **Find the minimum possible value of $2x^2 - 3x + 7$?**

a) $\frac{47}{8}$ b) $\frac{3}{2}$ c) $\frac{3}{4}$ d) $\frac{3}{8}$

3. **What maximum value can be attained by the expression $-x^2 + 7x - 12$, where $x \epsilon R$.**

a) 4 b) -4 c) $\frac{1}{4}$ d) $-\frac{1}{4}$

④ **Find the maximum value of the expression** $\frac{1}{x^2+5x+10}$

a) $\frac{15}{2}$ b) 1 c) $\frac{4}{15}$ d) 2

⑤ For what value of 'k', does the equation $x^2 + kx + 125 = 100$ has unique positive root?

a) 10 b) – 10 c) 10 or – 10 d) None

6 **What is the maximum integer that 'k' can take such that the roots of the equation $2x^2 + 5x + k$ are real?**

a) 3 **b) 4** **c) – 3** **d) 0**

7 **What is the smallest integer value of ' k' such that the roots of the equation $x^2 - 5x + k + 2 = 0$ are imaginary?**

a) – 5 **b) 4** **c) 5** **d) none**

(2) A root of equation $ax^2 + bx + c = 0$ (where a, b and c are rational numbers) is $5 + 3\sqrt{3}$. What is the value of $\frac{a^2+b^2+c^2}{ab+bc+ca}$?

a) $\frac{35}{3}$ b) $\frac{37}{3}$ c) $-\frac{105}{8}$ d) $\frac{105}{8}$

9 If $x = \cfrac{2}{2+\cfrac{2}{2+\cfrac{2}{2+\cfrac{2}{\ddots\infty}}}}$ then what is the value of x ?

$$2+\cfrac{2}{2+\cfrac{2}{2+\cfrac{2}{\ddots\infty}}}$$

a) $-2 \pm 2\sqrt{3}$ b) Both c & d

c) $\sqrt{3} - 1$ d) $-\sqrt{3} - 1$

Factor/remainder theorem:

$$x^3 - 2x^2 - 5x + 6$$

10 When $f(x) = 15x^3 - 14x^2 - 4x + 10$ is divided by $(3x + 2)$, then the remainder is:

a) 2 b) -1 c) -2 d) 1

11 If $px^3 + x^2 + 3x + q$ is exactly divisible by $(x + 2)$ and $(x - 2)$, then the value of p and q are:

a) $p = -\frac{3}{4}$ and $q = 4$

b) $p = \frac{3}{4}$ and $q = 4$

c) $p = \frac{3}{4}$ and $q = -4$

d) $p = -\frac{3}{4}$ and $q = -4$

12 If $(x+2)$ and $(x-1)$ are the factors of (x^3+10x^2+mx+n), the values of m and n are:

a) $m=5, n=-3$ b) $m=17, n=-8$

c) $m=7, n=-18$ d) $m=23, n=19$

13 If $x^3 + 5x^2 + 10k$ leaves remainder $-2x$ when divided by $x^2 + 2$, then the value of k is:

a) -2 b) -1 c) 1 d) 2

(14) If $x^{40} + 30$ is divided by $x^4 + 1$, find the remainder

a) 30 b) 31 c) 29 d) 0

15 If $(x+1)$ and $(x-1)$ are factor of ax^3+bx^2+3x+5. Find the value of $a-2b$.

a) 1 b) -13 c) 7 d) 0

16 If $x^4 - 2x^3 + 3x^2 - ax + b$ is divided by $x - 1$ and $x + 1$ the remainders are 5 and 19 respectively. Find the remainder when this polynomial is divided by $x - 2$.

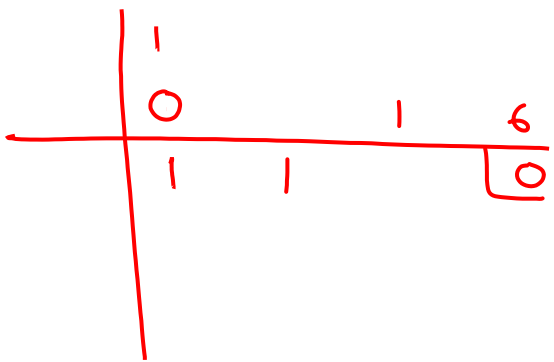
a) 10 b) 12 c) -1 d) 8

10

Higher degree equations:

Representation on graph:

Synthetic and long division:



17 **What is $(x-a)(x-b)(x-c)$ equal to?**

a) $x \ -(a+b+c)x \ +(bc+ca+ab)x-abc$

b) $x^3+(a+b+c)x^2+(bc+ca+ab)x+abc$

c) $x^3-(bc+ca+a)x^2+(a+b+c)x-abc$

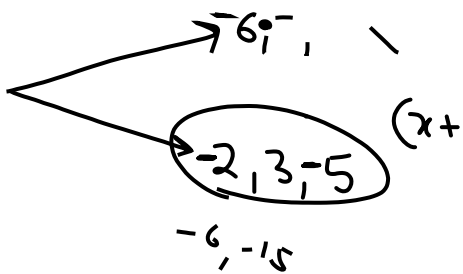
d) $x^3+(bc+ca+a)x^2-(a+b+c)x-abc$

13 If α, β and γ are the zeros of the polynomial $f(x) = ax^3 + bx^2 + cx + d$, then $\alpha^2 + \beta^2 + \gamma^2$ is equal to

a) $\frac{b\ -ac}{a^2}$ b) $\frac{b\ -2ac}{a}$ c) $\frac{b\ +2ac}{b^2}$ d) $\frac{b\ -2ac}{a^2}$

19 **What are the factors of $x^3 + 4x^2 - 11x - 30$?**

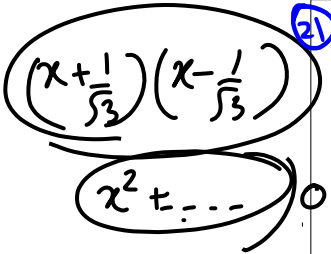
a) $(x-2), (x+3)$ and $(x+5)$
b) $(x+2), (x+3)$ and $(x-5)$
c) $(x+2), (x-3)$ and $(x+5)$
d) $(x+2), (x-3)$ and $(x-5)$



29 If x, y, z are the three factors of $a^3 - 7a - 6$, then the value of x + y + z will be

?

a) 3a b) 3 c) 6 d) a



21) Find the rational root of polynomial $f(x) = 3x^4 - 15x^3 + 17x^2 + 5x - 6$ if one roots is $-\frac{1}{3}$

a) -3, 2 b) 3, 2 c) 3,-2 d) -3, -2